(FP2) 5. Complex numbers

NO. 1

a. Use de Moivre's theorem to show that $\cot 6\theta = \dfrac{\cot^6\theta - 15\cot^4\theta + 15\cot^2\theta - 1}{6\cot^5\theta - 20\cot^3\theta + 6\cot\theta}$. [6]

b. Hence obtain the roots of the equation

$$x^6 - 6x^5 - 15x^4 + 20x^3 + 15x^2 - 6x - 1 = 0$$

in the form $\cot(q\pi)$, where q is a rational number. [4]

NO. 2

Find the roots of the equation $z^3 = -108\sqrt{3} + 108\mathrm{i}$, giving your answers in the form $r(\cos\theta + \mathrm{i}\sin\theta)$, where $r > 0$ and $0 < \theta < 2\pi$. [5]

NO. 3

a. Use de Moivre's theorem to show that

$$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta.$$ [4]

b. Hence obtain the roots of the equation

$$32x^5 - 40x^3 + 10x - \sqrt{2} = 0$$

in the form $\cos(q\pi)$, where q is a rational number. [4]

NO. 4

Find the roots of the equation $(z+5\mathrm{i})^3 = 4+4\sqrt{3}\,\mathrm{i}$, giving your answers in the form $r\cos\theta$ + i($r\cos\theta$ – 5), where r > 0 and 0 < θ < 2π. [5]

NO. 5

a. By considering the binomial expansion of $(z+z^{-1})^4$, where $z = \cos\theta + i\sin\theta$, use de Moivre's theorem to show that $\cos^4\theta = \frac{1}{8}(\cos 4\theta + 4\cos 2\theta + 3)$. [5]

b. Use the substitution $x = \sin\theta$ to find the exact value of $\int_0^{\frac{1}{2}} (1-x^2)^{\frac{3}{2}}\,dx$. [3]

NO. 6

a. Write down the fourth roots of unity. [1]

b. Use de Moivre's theorem to show that $\cos 4\theta = 8\cos 4\theta - 8\cos 2\theta + 1$. [4]

c. Hence obtain the real roots of the equation

$$16(8x^4 - 8x^2 + 1)^4 - 9 = 0$$

in the form $\cos(q\pi)$, where q is a rational number. [5]

NO. 7

a. Use de Moivre's theorem to show that

$$\operatorname{cosec} 7\theta = \frac{\operatorname{cosec}^7\theta}{7\operatorname{cosec}^6\theta - 56\operatorname{cosec}^4\theta + 112\operatorname{cosec}^2\theta - 64}. \quad [6]$$

b. Hence obtain the roots of the equation

$$x^7 - 14x^6 + 112x^4 - 224x^2 + 128 = 0$$

in the form cosec $q\pi$, where q is rational. [5]